Subject: FLUID MECHANICS & HYDRAULIC MACHINES LAB

Subject Code: 23A01506

LABORATORY MANUAL

Fluid Mechanics and Hydraulic Machines Lab (B.Tech Civil, JNTUA R23)

Course Objectives

- Understand the principles of fluid mechanics and validate fundamental concepts through experiments.
- Determine discharge coefficients for various flow measurement devices and analyse flow behaviour.
- Evaluate energy losses in pipes, open channels and hydraulic jumps to improve flow efficiency.
- Analyse the impact of jet forces on vanes and their applications in hydraulic machines.
- Assess the performance characteristics of hydraulic turbines and pumps under different operating conditions.

Course Outcomes

- Verify Bernoulli's equation and apply it to real-life fluid flow problems.
- Determine the coefficient of discharge for orifices, notches and flow meters.
- Determine head losses due to friction and minor losses in pipe flow systems.
- Analyse the impact of jets on vanes and its significance in hydraulic machinery.
- Assess the performance of turbines and pumps under different conditions and recommend optimal operating parameters.

Note minimum 10 out of the 13 experiments are to be conducted. All 13 are given here. Take g = 9.81 m/s2 and density of water = 1000 kg/m3.

List of Experiments

No	Experiment
1	Verification of Bernoulli's Equation
2	Coefficient of Discharge of a Small Orifice by Constant Head Method
3	Calibration of Venturimeter / Orifice Meter
4	Calibration of Triangular / Rectangular / Trapezoidal Notch
5	Determination of Minor Losses in Pipe Flow
6	Determination of Friction Factor of a Pipeline
7	Determination of Energy Loss in Hydraulic Jump
8	Determination of Manning's and Chezy's Constants for Open Channel Flow
9	Impact of Jet on Vanes
10	Performance Characteristics of Pelton Wheel Turbine
11	Performance Characteristics of Francis Turbine
12	Performance Characteristics of Kaplan Turbine
13	Performance Characteristics of a Single Stage / Multistage Centrifugal Pump

Experiment 1 – Verification of Bernoulli's Equation

Aim

To verify Bernoulli's theorem for the flow of water through a tapering duct (variable area conduit).

Apparatus

- Bernoulli's apparatus (tapering duct with 6 piezometer tubes), inlet and outlet valves, supply tank with pump, collecting (measuring) tank with piezometer scale, stop watch, scale.

Theory

For steady, incompressible, frictionless flow the total energy per unit weight along a streamline is constant.

Bernoulli equation p1/w + v1 squared/2g + z1 = p2/w + v2 squared/2g + z2 = constant. For a horizontal duct the piezometric head h plus the velocity head v squared/2g is the same at all sections. Discharge Q = Volume / time, velocity v = Q / A.

Procedure

- Fill the supply tank and start the pump; adjust the inlet valve so that water flows steadily and all piezometers show steady levels.
- Note the piezometer readings (h) at the 6 sections.
- Measure the discharge by closing the drain of the measuring tank and noting the time for a collection of 10 litres.
- Compute velocity at each section and the total head; repeat for another discharge if required.

Observations

Volume collected = 10 litres in t = 24.0 s. Discharge Q = 0.417 litre/s = 4.167e-04 m3/s.

Section	Dia (mm)	Area (cm2)	Velocity v (m/s)	v2/2g (cm)	Piezometer h (cm)	Total head h + v2/2g (cm)
1	25	4.91	0.85	3.7	38.3	42.0
2	20	3.14	1.33	9.0	32.8	41.8
3	16	2.01	2.07	21.9	19.6	41.5
4	12.5	1.23	3.40	58.8	-17.7	41.1
5	16	2.01	2.07	21.9	18.9	40.8
6	20	3.14	1.33	9.0	31.4	40.4

Result and Precautions

- The total head is almost constant at all sections (small fall of about 1.6 cm is due to friction loss), so Bernoulli's theorem is verified.
- Remove air bubbles from the piezometers; take readings only when the levels are steady; avoid parallax error.

Experiment 2 - Coefficient of Discharge of a Small Orifice (Constant Hea

Aim

To determine the coefficient of discharge (Cd) of a small circular orifice by the constant head method.

Apparatus

- Constant head tank with a circular orifice (diameter 12 mm), overflow arrangement, hook gauge or scale for head, measuring tank, stop watch.

Theory

A jet flowing from a small orifice under head H has the theoretical velocity v = sqrt(2 g H) (Torricelli). Actual discharge is less because of contraction of the jet (vena contracta) and friction.

Formulas Q theoretical = a x sqrt(2 g H), where a = area of orifice = pi d squared / 4. Q actual = Volume / time. Coefficient of discharge Cd = Q actual / Q theoretical = Cc x Cv.

Procedure

- Start the pump and keep the head in the supply tank constant using the overflow.
- Note the head H above the centre of the orifice with the scale.
- Collect 10 litres of water in the measuring tank and note the time.
- Repeat for different heads and find Cd for each.

Observations

Diameter of orifice d = 12 mm, area a = 1.131 cm2.

No	Head H (cm)	Volume V (litre)	Time t (s)	Q actual (cm3/s)	Q theoretical (cm3/s)	Cd
1	40	10	51.6	193.8	316.8	0.612
2	35	10	55.0	181.8	296.4	0.613
3	30	10	58.5	170.9	274.4	0.623
4	25	10	65.3	153.1	250.5	0.611

Result and Precautions

- Mean Cd of the orifice = 0.615 (the usual value is 0.60 to 0.62).
- Keep the head constant while collecting water; read the head at the level of the orifice centre; make sure the orifice edge is sharp and clean.

Experiment 3 – Calibration of Venturimeter / Orifice Meter

Aim

To calibrate a venturimeter (and orifice meter) and to determine its coefficient of discharge.

Apparatus

- Pipe line with venturimeter (inlet diameter 30 mm, throat diameter 15 mm), mercury differential U-tube manometer, flow control valve, pump and sump, measuring tank, stop watch.

Theory

Venturimeter works on Bernoulli's principle; the pressure falls at the throat and the pressure difference measures the discharge.

Formulas Q theoretical = [a1 a2 / sqrt(a1 squared - a2 squared)] x sqrt(2 g h). Head difference h = x (Sm / Sw - 1) = 12.6 x for mercury manometer. Cd = Q actual / Q theoretical. For an orifice meter the same method is used and Cd is about 0.60 to 0.65.

Procedure

- Note the pipe and throat diameters and the manometer fluid.
- Start the pump, open the control valve and allow steady flow; remove air from the manometer lines.
- Note the manometer readings and collect 10 litres in the measuring tank noting the time.
- Repeat for 4 or 5 different flow rates.
- Plot Q actual against sqrt(h); the slope of the straight line is Cd times the meter constant.

Observations

a1 = 7.069 cm2, a2 = 1.767 cm2.

No	Manometer reading x (cm)	Head h = 12.6 x (m of water)	Volume (litre)	Time t (s)	Q actual (m3/s x10-4)	Q theoretical (m3/s x10-4)	Cd
1	6	0.756	10	14.7	6.80	7.03	0.968
2	8	1.008	10	12.7	7.87	8.12	0.970
3	10	1.260	10	11.5	8.70	9.07	0.958
4	12	1.512	10	10.4	9.62	9.94	0.967

Result and Precautions

- Mean Cd of venturimeter = 0.966.
- Remove air bubbles from the pipe and manometer; take readings after steady flow; avoid very low flows.

Experiment 4 - Calibration of Rectangular / Triangular / Trapezoidal Notch

Aim

To find the coefficient of discharge of rectangular and triangular (V) notches.

Apparatus

- Notch tank with rectangular notch (width b = 10 cm) and V-notch (90 degree), hook gauge, stilling baffle, measuring tank, stop watch, pump and sump.

Theory

A notch is an opening in the side of a tank through which water flows with a free surface; H is the head over the crest (or vertex).

Rectangular notch Q theoretical = (2/3) b sqrt(2g) H to the power 3/2. Triangular notch :: Q theoretical = (8/15) tan(theta/2) sqrt(2g) H to the power 5/2 (for 90 degree notch tan(theta/2) = 1). Trapezoidal notch :: Q = rectangular part + 2 triangular parts. Cd = Q actual / Q theoretical.

Procedure

- Note the crest level of the notch with the hook gauge when the water just touches the crest (zero reading).
- Admit water and let the level become steady; read the level with the hook gauge and subtract the zero reading to get H.
- Collect water in the measuring tank for a measured time and find Q actual.
- Repeat for different heads; repeat the test for the other notch.

Observations - Rectangular notch (b = 10 cm)

No	Head H (cm)	Volume (litre)	Time t (s)	Q actual (cm3/s)	Q theoretical (cm3/s)	Cd
1	3.0	5	5.2	961.5	1534.4	0.627
2	3.5	5	4.1	1219.5	1933.6	0.631
3	4.0	5	3.5	1428.6	2362.4	0.605

4	4.5	5	2.9	1724.1	2818.9	0.612

Observations – Triangular notch (90 degree)

No	Head H (cm)	Volume (litre)	Time t (s)	Q actual (cm3/s)	Q theoretical (cm3/s)	Cd
1	4.0	5	11.3	442.5	756.0	0.585
2	5.0	5	6.2	806.5	1320.6	0.611
3	6.0	5	3.9	1282.1	2083.2	0.615
4	7.0	5	2.8	1785.7	3062.6	0.583

Result and Precautions

- Mean Cd of rectangular notch = 0.618; mean Cd of triangular notch = 0.599.
- Wait for steady head; avoid disturbance of the water surface near the hook gauge; the crest must be sharp and the water level just behind the baffle.

Experiment 5 – Determination of Minor Losses in Pipe Flow

Aim

To determine the loss of head and the loss coefficient K due to sudden enlargement (and other fittings) in a pipe.

Apparatus

- Pipe set up with sudden enlargement from 20 mm to 40 mm diameter, bend, elbow and contraction fittings, piezometers (or pressure gauges) before and after the fitting, measuring tank, stop watch, scale.

Theory

Minor losses are caused by changes in the size or direction of flow (enlargement, contraction, bends, valves). They are expressed in terms of the velocity head.

Formulas Loss in sudden enlargement hL = (v1 - v2) squared / 2g. In general hL = K v squared / 2g. From Bernoulli equation the experimental loss hL = (h1 - h2) + (v1 squared - v2 squared)/2g, where h1 and h2 are piezometer heads before and after the enlargement.

Procedure

- Note the diameters before and after the fitting.
- Start the flow and adjust the valve to get a steady discharge; remove air from the piezometer tubes.
- Note the piezometer readings h1 and h2 and find the discharge by volume and time.
- Repeat for different discharges; compute hL and K; repeat for other fittings (bend, contraction) in the same way.

Observations

d1 = 20 mm, d2 = 40 mm, a1 = 3.142 cm2, a2 = 12.566 cm2. Volume collected = 10 litres in each trial.

No	Time t (s)	Q (litre/s)	v1 (m/s)	v2 (m/s)	h1 (cm)	h2 (cm)	hL experimental (cm)	hL theory (cm)	K = hL / (v1 squared/2g)
1	18.2	0.55	1.75	0.44	62.7	68.0	9.32	8.77	0.60
2	13.3	0.75	2.39	0.60	62.9	73.5	16.77	16.42	0.57
3	10.5	0.95	3.03	0.76	61.1	79.7	25.31	26.35	0.54

Result and Precautions

- Experimental loss of head is close to the theoretical value (Borda-Carnot); mean K = 0.57.
- Pressure rises after a sudden enlargement, so h2 is more than h1 (a negative difference shows this). Avoid air bubbles; take readings only when steady.

Experiment 6 – Determination of Friction Factor of a Pipeline

Aim

To determine the Darcy friction factor f of a pipe for different Reynolds numbers.

Apparatus

- Pipe (G.I. or PVC) of diameter 20 mm with pressure tappings 2.0 m apart, mercury U-tube differential manometer, control valve, sump and pump, measuring tank, stop watch.

Theory

Head loss due to friction in a pipe is given by the Darcy-Weisbach equation.

Formulas hf = f L v squared / (2 g d), so f = 2 g d hf / (L v squared). hf = x (Sm/Sw - 1) = 12.6 x for a mercury manometer. Reynolds number Re = v d / nu (nu = 1 x 10 to the power -6 m2/s for water). For laminar flow f = 64/Re; for smooth turbulent pipes f = 0.316 / Re to the power 0.25 (Blasius, Re up to 100000).

Procedure

- Note the pipe diameter, length between tappings and the manometer liquid.
- Start the pump, open the valve fully and remove air from the pressure lines.
- Note the manometer reading x and find the discharge by collecting 10 litres of water in the measuring tank.
- Reduce the flow in steps and repeat for 4 to 5 readings; compute f and Re.
- Plot f against Re (log scale).

Observations

Diameter d = 20 mm, length L = 2.0 m, area a = 3.142 cm2.

No	Volume (litre)	Time t (s)	Q (m3/s x10-4)	v (m/s)	Manometer x (cm)	hf = 12.6 x (m)	f = 2 g d hf / (L v squared)	Re = v d / nu
1	10	26.5	3.77	1.20	1.8	0.227	0.0308	24023
2	10	17.7	5.65	1.80	3.6	0.454	0.0275	35967

3	10	13.3	7.52	2.39	5.9	0.743	0.0255	47866

Result and Precautions

- The friction factor is between 0.0255 and 0.0308; it decreases as Re increases (turbulent flow in a rough pipe).
- Clear air bubbles from the manometer; take the readings when the flow is steady; keep tappings away from bends and valves.

Experiment 7 – Determination of Energy Loss in Hydraulic Jump

Aim

To study the hydraulic jump in a rectangular channel and to determine the energy loss and the conjugate depths.

Apparatus

- Rectangular tilting flume (width 10 cm) with a sluice gate at the entry and an adjustable tail gate, point gauges or scale, measuring tank, stop watch.

Theory

A hydraulic jump is the sudden rise of the water surface when a supercritical flow (Fr greater than 1) changes into subcritical flow; a large amount of energy is lost as turbulence.

Formulas Froude number Fr1 = v1 / sqrt(g y1). Conjugate depth y2 = (y1/2) x [sqrt(1 + 8 Fr1 squared) - 1]. Energy loss EL = (y2 - y1) cubed / (4 y1 y2). Specific energy E1 = y1 + v1 squared/2g. Length of jump is about 6 y2.

Procedure

- Fix the sluice gate opening and start the flow; adjust the tail gate until a steady jump forms in the channel.
- Measure the depths y1 (before the jump) and y2 (after the jump) and the discharge Q by volumetric method.
- Calculate v1, Fr1, theoretical y2 and the energy loss; repeat for other discharges

or gate openings.

Observations

Channel width b = 10 cm.

No	Q (m3/s x10-3)	y1 (cm)	y2 (cm)	v1 (m/s)	Fr1	y2 theory (cm)	Energy loss EL (cm)	EL/E1 (%)
1	4.0	1.5	14.0	2.67	6.95	14.0	23.25	61.6
2	4.5	1.65	14.6	2.73	6.78	15.0	22.54	57.0
3	5.0	1.8	15.7	2.78	6.61	16.0	23.76	57.8

Result and Precautions

- The measured conjugate depths agree with the theoretical values; the energy dissipated by the jump is 61.6 to 57.8 percent of the incoming energy.
- Measure y1 just before the jump and y2 after the roller; keep the jump stable; use a sharp pointed gauge.

Experiment 8 - Manning's and Chezy's Constants for Open Channel Flow

Aim

To determine Chezy's constant C and Manning's roughness coefficient n for flow in a rectangular open channel.

Apparatus

- Rectangular tilting flume (width 15 cm, length 5 m) with a bed slope arrangement, point gauge, inlet valve, measuring tank, stop watch, scale.

Theory

For uniform flow in an open channel the velocity is given by Chezy's and Manning's formulas.

Formulas Chezy v = C sqrt(R S). Manning v = (1/n) R to the power 2/3 x S to the power 1/2. Hydraulic radius R = A/P, A = b y, P = b + 2y. C = (1/n) R to the power 1/6. S is the bed slope.

Procedure

- Set the flume to a bed slope of 1 in 400 (S = 0.0025).
- Admit water and adjust until uniform flow (constant depth along the length) is obtained.
- Measure the depth y with the point gauge at three places and take the mean.
- Find the discharge by collecting water in the measuring tank; compute v, R, C and n.
- Repeat for different discharges.

Observations

Width b = 15 cm, bed slope S = 1/400 = 0.0025.

No	Depth y (cm)	Area A (cm2)	Wetted perimeter P (cm)	R = A/P (cm)	Volume (litre)	Time t (s)	Q (m3/s x10-3)	v (m/s)	Chezy C	Manning n
1	5.0	75.0	25.0	3.00	20	5.8	3.45	0.460	53.1	0.0105
2	6.5	97.5	28.0	3.48	20	4.3	4.65	0.477	51.1	0.0112
3	8.0	120.0	31.0	3.87	20	3.1	6.45	0.538	54.7	0.0106

Result and Precautions

- Mean Chezy's constant C = 53.0 m to the power 1/2 per second. Mean Manning's n = 0.0108 (about 0.010 for a smooth perspex or glass channel).
- Make sure the flow is uniform before measuring; measure the depth away from the entry and exit; check the bed slope.

Experiment 9 – Impact of Jet on Vanes

Aim

To determine the force exerted by a jet of water on flat and curved vanes and to verify the momentum equation.

Apparatus

- Jet impact apparatus (transparent cylindrical chamber) with nozzle 10 mm diameter, set of vanes (flat plate, hemispherical cup, inclined and curved vane), lever with weight pan and balancing weights, supply pump and sump, measuring tank, stop watch.

Theory

By Newton's second law the force on the vane equals the rate of change of momentum of the jet.

Formulas Jet velocity v = Q / a. Force on a flat plate normal to the jet F = rho a v squared. On a hemispherical vane (jet turned through 180 degree) F = 2 rho a v squared. On a curved vane F = rho a v squared (1 + cos(phi)). Actual force F = W g (weight on the pan, with lever arm ratio 1). Efficiency of the vane = F actual / F theoretical.

Procedure

- Fit the flat plate vane and balance the lever arm with the pointer at zero.
- Start the pump and allow the jet to strike the vane; add weights on the pan till the lever returns to the zero position.
- Note the weight W and find the discharge by volume and time method.
- Repeat for another discharge and then repeat the same for the hemispherical (and other) vanes.

Observations

Nozzle diameter = 10 mm, area a = 0.785 cm2.

Vane	Q (litre/s)	Jet velocity v (m/s)	F theoretical (N)	Weight W (g)	F actual = W g (N)	k = F actual / F theory
Flat plate (90 deg)	0.298	3.80	1.134	104	1.020	0.90
Flat plate (90 deg)	0.340	4.33	1.470	139	1.364	0.93
Hemispherical (180 deg)	0.302	3.85	2.323	222	2.178	0.94
Hemispherical (180 deg)	0.341	4.34	2.960	282	2.766	0.93

Result and Precautions

- The actual force is 90 to 97 percent of the theoretical force; the force on the hemispherical vane is about twice that on the flat plate (for the same jet). The small loss is due to friction and splash.
- Keep the jet central on the vane; let the lever come to rest before reading; check the zero of the lever.

Experiment 10 – Performance Characteristics of Pelton Wheel Turbine

Aim

To conduct a performance test on a Pelton wheel turbine at constant head and to draw its characteristic curves.

Apparatus

- Pelton wheel turbine (impulse turbine) with spear and nozzle, pressure gauge, supply pump, rope brake dynamometer with spring balances, tachometer, venturimeter or notch for discharge, stop watch.

Theory

The Pelton wheel is an impulse turbine for high head and low discharge; the jet of water from the nozzle strikes the double hemispherical buckets, and the flow is controlled by a spear. The pressure at the buckets is atmospheric.

Formulas Input (water) power = w Q H = rho g Q H. Torque T = (W1 - W2) g R, where R = (D + d)/2 is the effective radius of the brake drum. Brake power BP = 2 pi N T / 60. Overall efficiency = BP / input power x 100. Unit speed Nu = N / sqrt(H), unit power Pu = P / H to the power 3/2.

Procedure

- Check that the guide arrangement is closed, prime the pump and start it; open the supply valve slowly to the required head and note the pressure gauge reading.
- With no brake load note the no-load (runaway) speed. Apply the brake load gradually by increasing the spring balance load.
- For each load note the speed N, spring balance readings W1 and W2 and the discharge (from the venturimeter or the notch).
- Keep the head constant by adjusting the supply valve, and repeat for 5 to 6 loads up to the speed at which the turbine just stops.

Observations

Net head H = 30 m, brake drum diameter D = 0.30 m, rope diameter d = 0.02 m, effective radius R = 0.16 m. Pressure gauge reading = 3.0 kg/cm2 (about 30 m of water), constant head.

No	N (rpm)	Q (litre/s)	W1 (kg)	W2 (kg)	Torque T (N m)	Brake power BP (W)	Input power (W)	Efficiency (%)
1	300	5.12	14.6	0.8	21.66	680	1507	45.2
2	500	5.08	12.5	1.2	17.74	929	1495	62.1
3	700	5.15	10.5	1.1	14.75	1082	1516	71.4

4	900	5.12	7.4	0.7	10.52	991	1507	65.8
5	1100	5.07	4.9	0.6	6.75	777	1492	52.1

Result

- Maximum efficiency of the turbine = 71.4 % at N = 700 rpm. The efficiency is maximum near a speed of 0.45 times the jet velocity (bucket speed), as in theory.
- Plot the curves of BP, torque and efficiency against speed (main characteristic curves). Precaution: apply the brake load gradually; keep the net head constant; make sure the cooling water is flowing in the brake drum.

Experiment 11 – Performance Characteristics of Francis Turbine

Aim

To conduct a performance test on a Francis turbine at constant head and to draw its characteristic curves.

Apparatus

- Francis turbine (reaction turbine) with guide vanes, spiral casing and draft tube, pressure gauge and vacuum gauge, supply pump, rope brake dynamometer with spring balances, tachometer, venturimeter or notch, stop watch.

Theory

Francis turbine is a mixed flow reaction turbine for medium head. Water enters radially from the spiral casing through the guide vanes and leaves axially to the draft tube; the runner is always full of water and works under both pressure and kinetic energy. Net head H = pressure gauge head + vacuum gauge head + height difference.

Formulas Input (water) power = w Q H = rho g Q H. Torque T = (W1 - W2) g R, where R = (D + d)/2 is the effective radius of the brake drum. Brake power BP = 2 pi N T / 60. Overall efficiency = BP / input power x 100. Unit speed Nu = N / sqrt(H), unit power Pu = P / H to the power 3/2.

Procedure

- Check that the guide vanes are closed, prime the pump and start it; open the supply valve slowly to the required head and note the pressure gauge reading.
- With no brake load note the no-load (runaway) speed. Apply the brake load gradually by increasing the spring balance load.
- For each load note the speed N, spring balance readings W1 and W2 and the discharge (from the venturimeter or the notch).
- Keep the head constant by adjusting the supply valve, and repeat for 5 to 6 loads up to the speed at which the turbine just stops.

Observations

Net head H = 15 m, brake drum diameter D = 0.30 m, rope diameter d = 0.02 m, effective radius R = 0.16 m. Pressure gauge 1.2 kg/cm2, vacuum gauge 200 mm of Hg, net head 15 m.

No	N (rpm)	Q (litre/s)	W1 (kg)	W2 (kg)	Torque T (N m)	Brake power BP (W)	Input power (W)	Efficiency (%)
1	600	11.90	7.2	1.0	9.73	611	1751	34.9
2	800	11.98	8.1	1.1	10.99	920	1763	52.2
3	1000	11.97	8.1	1.2	10.83	1134	1761	64.4
4	1200	12.08	6.6	0.4	9.73	1223	1778	68.8
5	1400	11.93	5.7	1.1	7.22	1059	1755	60.3

Result

- Maximum efficiency of the turbine = 68.8 % at N = 1200 rpm. Francis turbines work well at 60 to 70 percent at partial gate opening and best near

the design speed.

- Plot the curves of BP, torque and efficiency against speed (main characteristic curves). Precaution: apply the brake load gradually; keep the net head constant; make sure the cooling water is flowing in the brake drum.

Experiment 12 - Performance Characteristics of Kaplan Turbine

Aim

To conduct a performance test on a Kaplan turbine at constant head and to draw its characteristic curves.

Apparatus

- Kaplan turbine (axial flow reaction turbine with adjustable runner blades and guide vanes), draft tube, pressure and vacuum gauges, supply pump, brake drum with spring balances, tachometer, venturimeter, stop watch.

Theory

The Kaplan turbine is an axial flow reaction turbine suitable for low head and large discharge. Both the guide vanes and the runner blades are adjustable, so high efficiency is kept even at part load (double regulation). The runner has few blades and a large hub (boss).

Formulas Input (water) power = w Q H = rho g Q H. Torque T = (W1 - W2) g R, where R = (D + d)/2 is the effective radius of the brake drum. Brake power BP = 2 pi N T / 60. Overall efficiency = BP / input power x 100. Unit speed Nu = N / sqrt(H), unit power Pu = P / H to the power 3/2.

Procedure

- Check that the guide vanes are closed, prime the pump and start it; open the supply valve slowly to the required head and note the pressure gauge reading.
- With no brake load note the no-load (runaway) speed. Apply the brake load gradually by increasing the spring balance load.

- For each load note the speed N, spring balance readings W1 and W2 and the discharge (from the venturimeter or the notch).
- Keep the head constant by adjusting the supply valve, and repeat for 5 to 6 loads up to the speed at which the turbine just stops.

Observations

Net head H = 6 m, brake drum diameter D = 0.30 m, rope diameter d = 0.02 m, effective radius R = 0.16 m. Guide vane opening = 75 %, constant head (pressure gauge plus vacuum gauge) = 6 m.

No	N (rpm)	Q (litre/s)	W1 (kg)	W2 (kg)	Torque T (N m)	Brake power BP (W)	Input power (W)	Efficiency (%)
1	400	19.99	8.7	1.2	11.77	493	1177	41.9
2	550	19.96	8.0	0.5	11.77	678	1175	57.7
3	700	20.05	8.0	1.0	10.99	805	1180	68.2
4	850	19.91	5.8	0.5	8.32	740	1172	63.2
5	1000	19.93	4.8	1.2	5.65	592	1173	50.4

Result

- Maximum efficiency of the turbine = 68.2 % at N = 700 rpm. Kaplan runner blades should be set at the correct angle for good efficiency at each load.
- Plot the curves of BP, torque and efficiency against speed (main characteristic curves). Precaution: apply the brake load gradually; keep the net head constant; make sure the cooling water is flowing in the brake drum.

Experiment 13 – Performance Characteristics of a Single Stage Centrifuga

Aim

To conduct a performance test on a centrifugal pump running at constant speed and to draw its characteristic curves.

Apparatus

- Centrifugal pump coupled to an electric motor, delivery pressure gauge, suction vacuum gauge, delivery valve, collecting (measuring) tank, energy meter (3200 rev/kWh), stop watch, tachometer.

Theory

A centrifugal pump raises the pressure of the liquid by the centrifugal action of its impeller. Efficiency is the ratio of the water power to the power supplied.

Formulas Total head H = pd x 10 + pv x 0.0136 + z, where pd is delivery pressure in kg/cm2 (1 kg/cm2 = 10 m of water), pv is suction vacuum in mm of Hg (1 mm Hg = 0.0136 m of water) and z = height between the gauges. Output (water) power = rho g Q H. Input power = (n / K) x (3600 / T) kW where n = revolutions of the energy meter disc in time T seconds and K = meter constant. Overall efficiency = output / input x 100.

Procedure

- Prime the pump and close the delivery valve fully; start the motor.
- Open the delivery valve to a required position and note the delivery pressure and the suction vacuum.
- Measure the discharge by collecting 10 litres in the collecting tank and note the time.
- Note the number of revolutions of the energy meter disc in 60 seconds.
- Repeat for different valve openings (5 or 6 readings); never run the pump for long with the delivery valve closed.
- Plot H, power input and efficiency against Q (main characteristic curves).

Observations

Height between gauges z = 0.4 m, energy meter constant = 3200 rev/kWh, constant speed N = 2900 rpm. Volume collected = 10 litres in each trial.

No	Delivery valve	Time t (s)	Q (litre/	Delivery gauge pd (kg/cm2)	Vacuum pv (mm Hg)	Head H (m)	Meter rev n (in 60 s)	Input power (W)	Output power (W)	Overall efficiency (%)
1	1/4 open	12.5	0.80	2.58	130	28.0	39	731	219	30.0
2	1/2 open	8.3	1.20	2.37	140	26.0	39	731	307	42.0
3	3/4 open	6.2	1.61	2.16	150	24.0	41	769	380	49.5
4	Nearly full	5.0	2.00	1.84	160	21.0	42	788	412	52.3
5	Fully open	4.2	2.38	1.43	170	17.0	45	844	397	47.1

Result

- Maximum overall efficiency = 52.3 % at a discharge of 2.00 litre/s and head 21.0 m (the best efficiency point).
- Head falls as discharge increases, while the efficiency rises up to a maximum and then falls. Precaution: prime the pump before starting; do not run with the delivery valve closed for long; take readings only when steady.

Note on multistage pump

- In a multistage pump several impellers are connected in series on one shaft; the total head is the sum of the heads of all stages while the discharge is the same. The test procedure is the same and the head is about n times that of one stage.

Summary of Important Formulas

Experiment	Key formula
Bernoulli	p/w + v squared/2g + z = constant
Orifice	Cd = Q actual / [a sqrt(2gH)]
Venturimeter	Q = Cd a1 a2 sqrt(2gh) / sqrt(a1 squared - a2 squared)
Rectangular notch	Q = (2/3) Cd b sqrt(2g) H to the power 3/2
Triangular notch	Q = (8/15) Cd tan(theta/2) sqrt(2g) H to the power 5/2
Minor loss	hL = K v squared/2g ; enlargement hL = (v1 - v2) squared/2g
Friction	hf = f L v squared/2g d
Hydraulic jump	y2 = (y1/2)(sqrt(1 + 8 Fr squared) - 1), EL = (y2 - y1) cubed/(4 y1 y2)
Chezy, Manning	v = C sqrt(RS) ; v = (1/n) R to the power 2/3 S to the power 1/2
Jet on vane	Flat plate F = rho a v squared ; hemispherical F = 2 rho a v squared
Turbine	BP = 2 pi N T / 60 ; efficiency = BP / (rho g Q H)
Pump	efficiency = rho g Q H / input power

Viva Voce Questions

- State Bernoulli's theorem and list its assumptions.
- What is the vena contracta? What is the value of Cc, Cv and Cd for an orifice?
- Why is a venturimeter preferred to an orifice meter? Why is Cd of the venturimeter higher?
- Differentiate between major and minor losses in pipes.
- What is Reynolds number? State the limits for laminar and turbulent flow.
- What is a hydraulic jump? Where is it used?
- Differentiate between impulse and reaction turbines. Give one example of each.
- Why does a Kaplan turbine have adjustable blades?

- What is priming of a centrifugal pump? What is cavitation?
- What do you understand by main characteristic and operating characteristic curves?